When the normalization procedure is completed (**QC & data transformation**), the **Visualization** module is automatically enabled. The **Visualization** module provides spatialGE users with functionality to generate high-quality figures showing the expression of genes in spatial context. Two options are available: **Quilt plot** and **Expression surface**.

Quilt plot

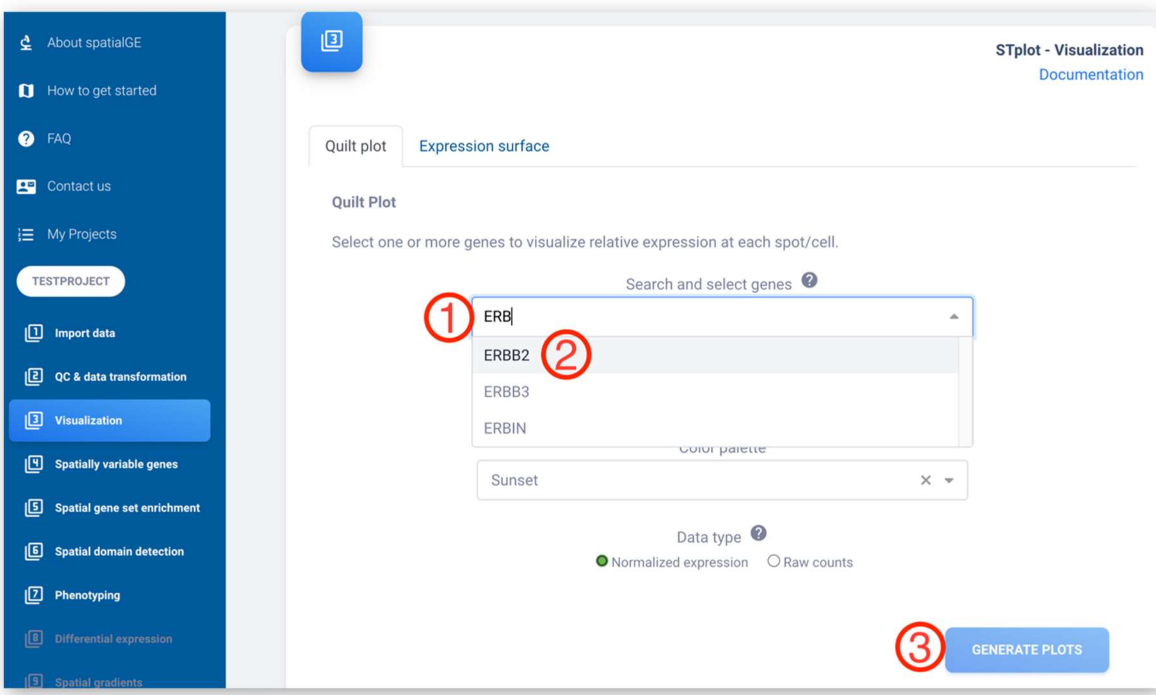
The quilt plots present each spot or cell as a point, colored according to the expression of one or more genes the user selects. It is a quick way to present spatial gene expression. The interface of **Quilt plot** has been designed to facilitate comparative analysis across samples.

To start, type the names of a few genes in the **Search and select genes** box. For this example, the gene names *ERBB2* (a.k.a. HER2) and *COL1A1* (a collagen gene) will be used. Notice that this textbox allows you to partially input a gene name and displays matching names for your convenience. When the gene name shows in the drop-down list, please click on it.

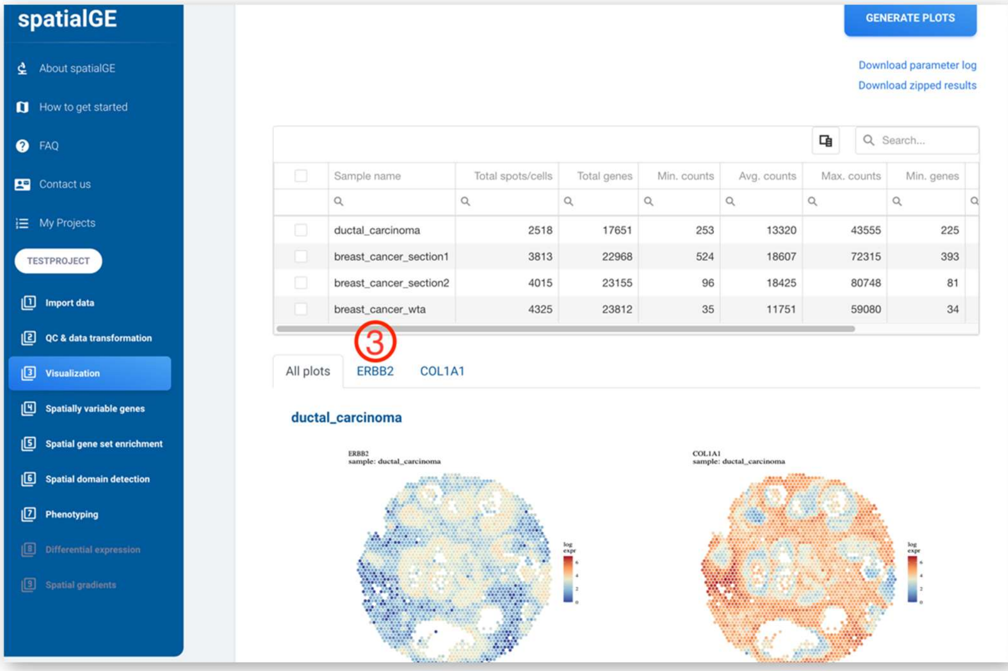
Note: The **Select and search genes** textbox is case-sensitive, for example ERBB2 is not the same as erbb2.

The **Point size** slider will be left at its default value (2), but the user can select smaller or larger dots to represent the spots/cells. The user also has the option to display the normalized or raw counts by selecting the appropriate option in **Data type**. Most users will probably want to plot the normalized expression, and for this example, the normalized will be plotted, too.

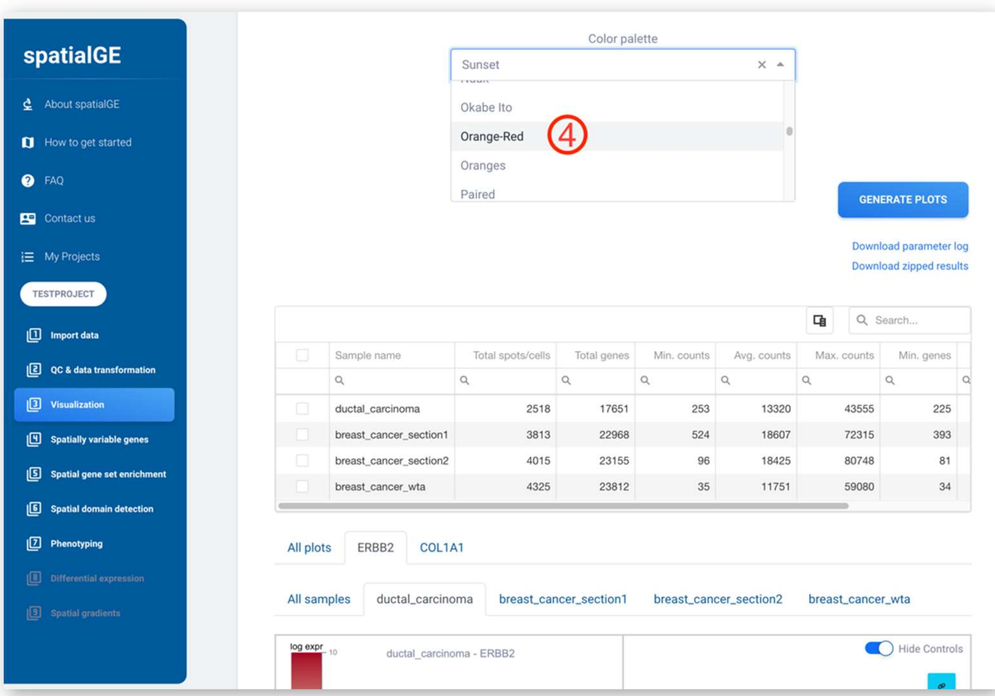
To make the plots, please click **GENERATE PLOTS**.



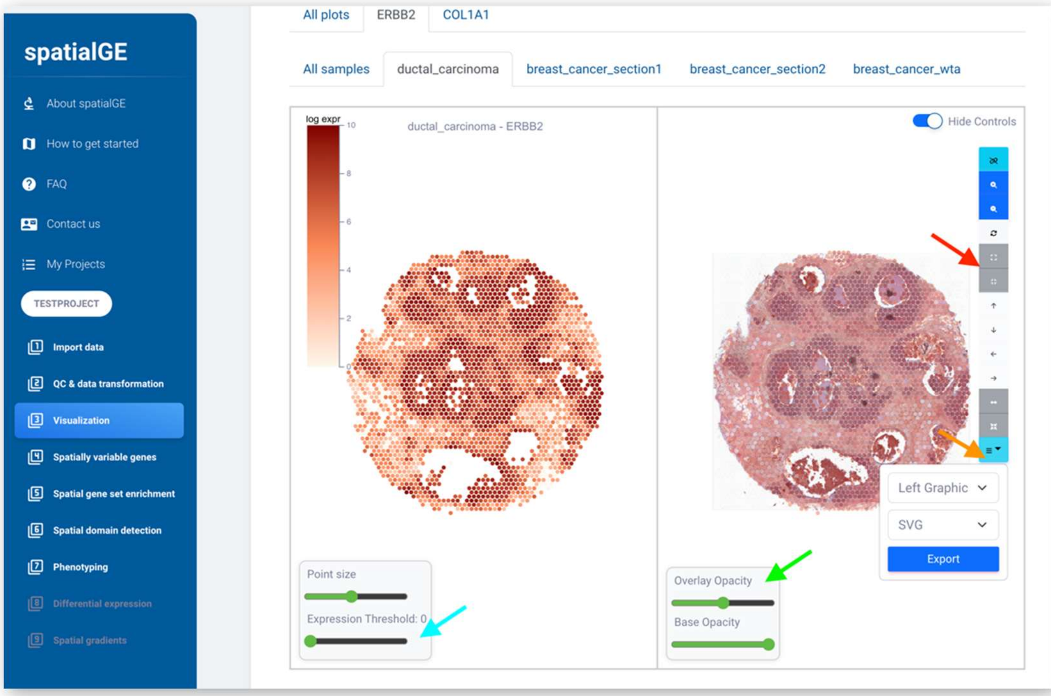
All the plots for all genes and samples are shown in the new **All plots** tab, but you can look at plots for each gene by clicking on one of the adjacent tabs. For example, please click the *ERRB2* tab.



Users can also choose a **Color palette** for the expression values at each spot/cell. For example, select the "Orange-Red" palette from the drop-down menu (the palettes are provided by the Khroma and RColorBrewer R packages).



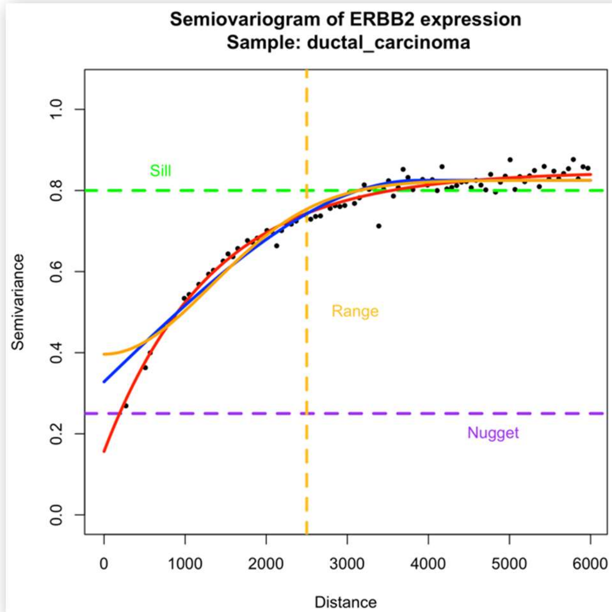
Color changes are reflected within each gene and sample tab. The plots can also be modified using several controls in the interface. Some of the most important controls include “Expression threshold” to hide spots/cells with gene expression under a user-selected level (turquoise arrow), controls for the opacity of the plot and tissue image (if available, green arrow), and adjustment of plot to tissue image (manual image registration, red arrow). Finally, images can be exported as **PDF/PNG/SVG** (orange arrow).



Expression surface plot

Some spatial transcriptomics platforms do not produce data at the single-cell level, and expression is measured at the “mini-bulk” level. Examples of these are the Visium samples in this documentation, in which each spot is approximately 55 microns in diameter, encompassing more than one cell. In addition, “mini-bulk” spatial transcriptomics do not sample each cell within the tissue but only those captured within the spots. In this case, it is possible to apply spatial statistics to estimate the expression values of the unsampled areas based on the sampled areas. This procedure is known as spatial interpolation and creates gene **expression surfaces**. In spatialGE the method known as “kriging” has been adapted to achieve this goal.

The generation of expression surfaces assumes that the expression values of two given ROIs/spots/cells are correlated by their distance. In other words, two spots that are closer are likely to have more similar expression values compared to two spots that are distant. This correlation between two given spots is often visualized as a semivariogram. Below, a semiovariogram of the expression of *ERBB2* in the “ductal_carcinoma” sample is shown. In this semiovariogram, it can be seen that the expression of *ERBB2* in this spatial transcriptomics sample is correlated to distance, and this correlation can be modeled using an exponential (red line), spherical (blue line), or gaussian (yellow line) covariance function.

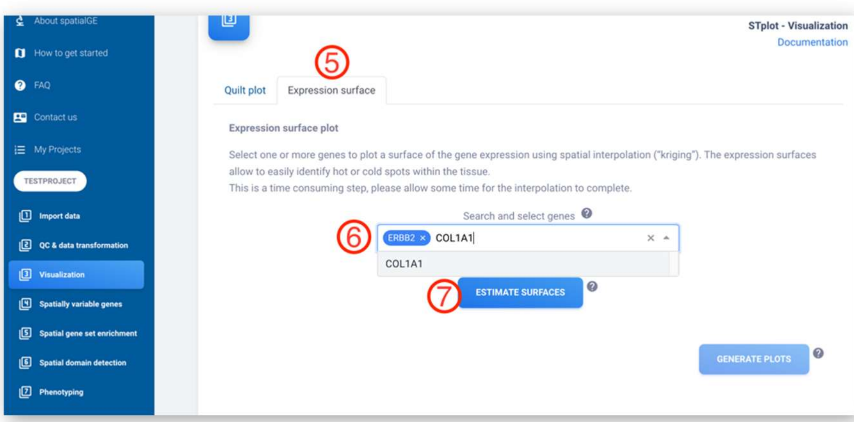


In spatialGE, the exponential function is the type of covariance function used to model the correlation of expression between spots.

To perform spatial interpolation on a set of genes, click the **Expression surface** tab. Please, begin by specifying a set of genes to generate expression surfaces. The genes to be interpolated are specified in the **Search and select genes** textbox. This textbox features auto-completion, and by typing a letter, a list of matching genes is shown. Please, enter ERBB2 in the **Search and select genes** and click on the gene name. Then enter the COL1A1 gene and click on the name. You are encouraged to add more genes, but please keep in mind that kriging is a computationally intensive method, and the more genes you add to the list, the more time it takes to complete.

Note: The **Select and search genes** textbox is case-sensitive, for example ERBB2 is not the same as erbb2.

To start the estimation of expression surfaces, please click the **ESTIMATE SURFACES** button.



Immediately after the estimation of surfaces is completed, the expression surface plots are shown below. The plots are generated with the "Sunset" color palette, but users can change the colors by selecting one of the options in the **Color palette** drop-down (purple arrow) followed by the **GENERATE PLOTS** button (pink arrow). The palettes are provided by the Khroma and RColorBrewer R packages, please refer to their documentation to learn more about other palettes. Because gene expression is a continuous measure, the diverging and sequential palettes are recommended.

The gene expression surfaces allow for a better exploration of tissue heterogeneity. In the example below, it can be seen that expression of the gene *COL1A1* resembles well the tissue architecture evident in the stained tissue image. The behavior of the interface is similar to that of **Quilt plots**, wherein plots for each gene have a dedicated tab, with sub-tabs for each sample. To display the expression surface next to the tissue image (if available), please go to the sub-tab specific to the sample to be observed. Users can also download expression surface plots in different formats by clicking the corresponding button (**PDF**/**PNG**/**SVG**, yellow arrow).